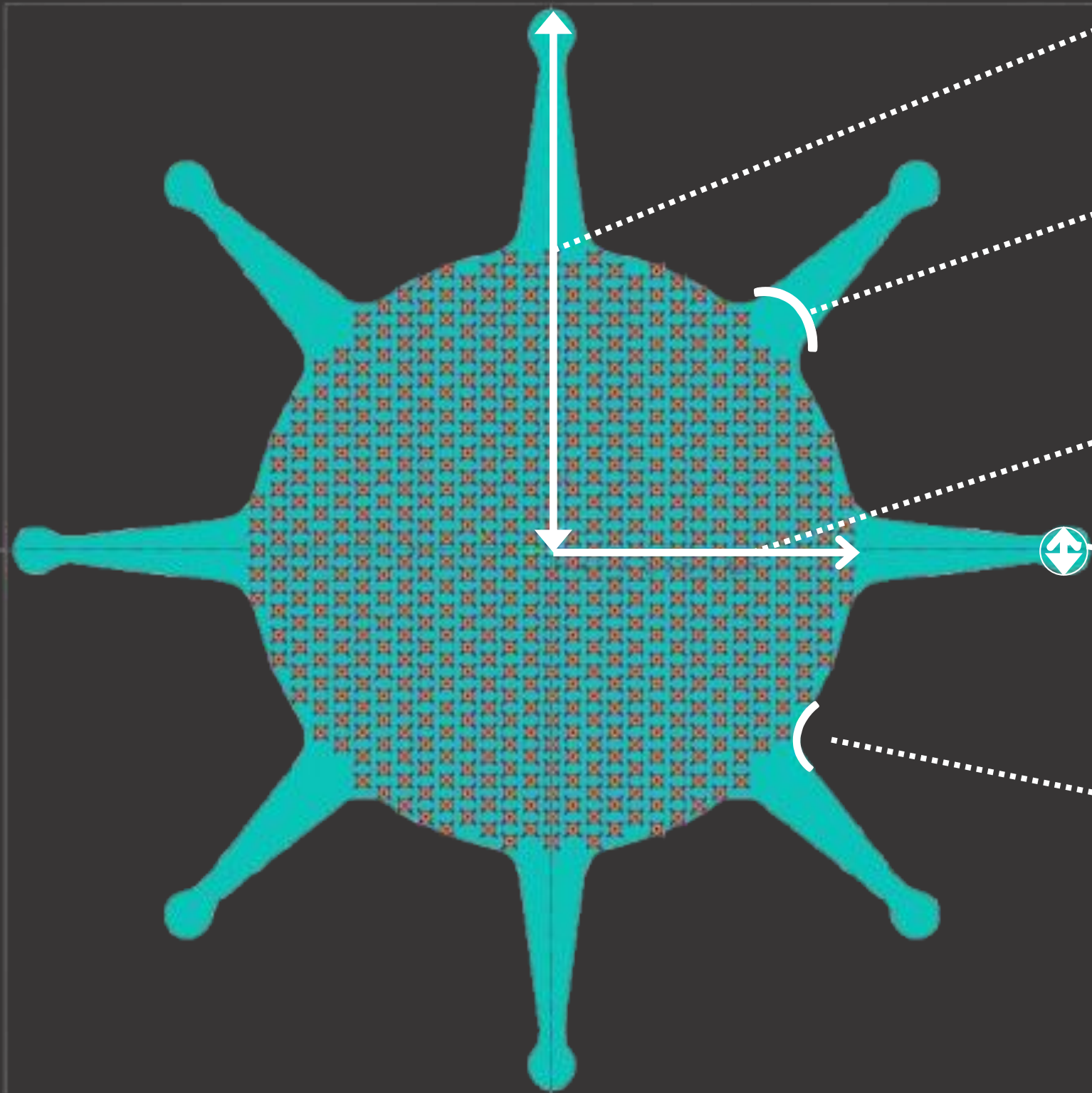


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# DEFORMABLE MIRROR

# PARAMETERS

## DEFORMABLE MIRROR



Arm Length

Arm Angle

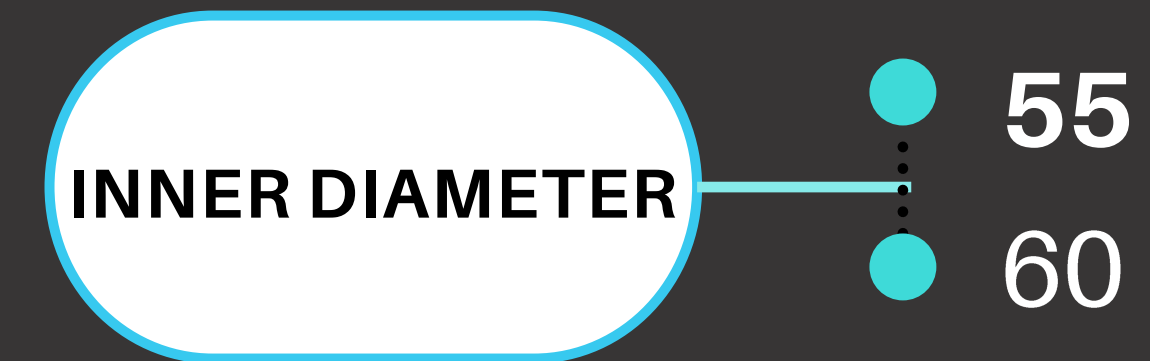
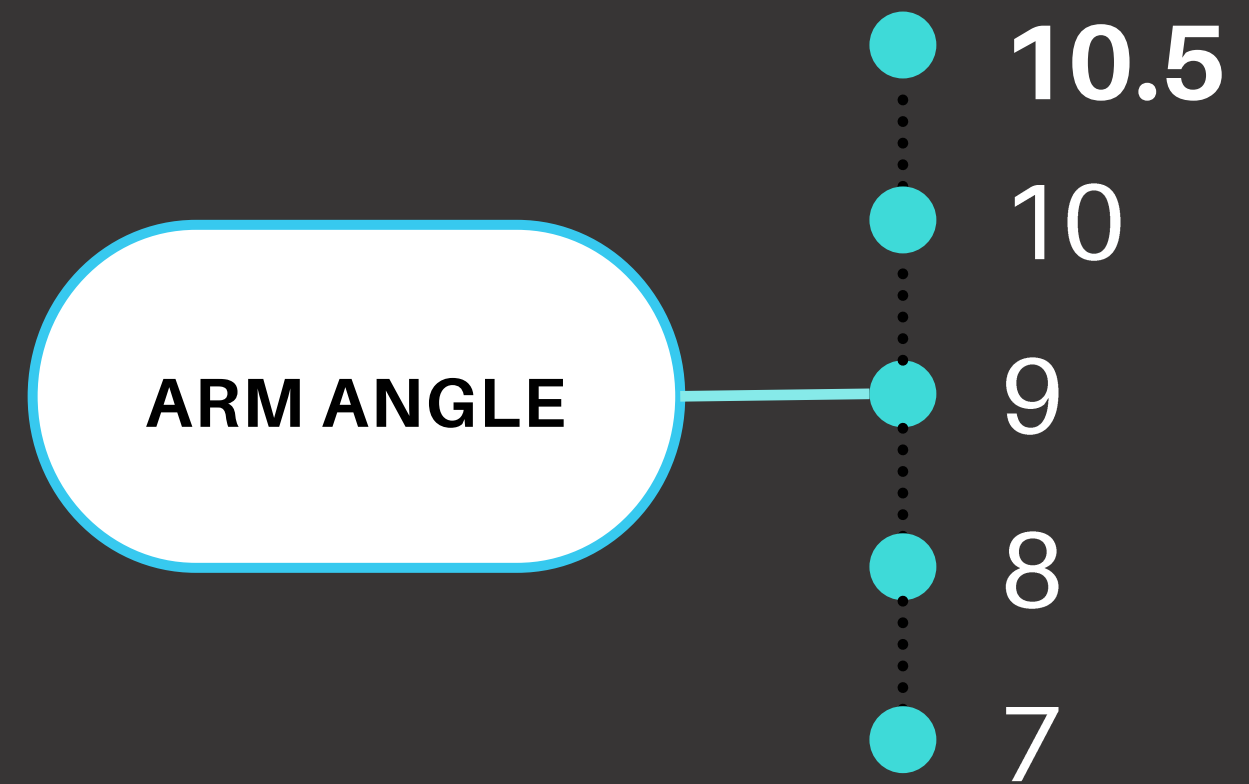
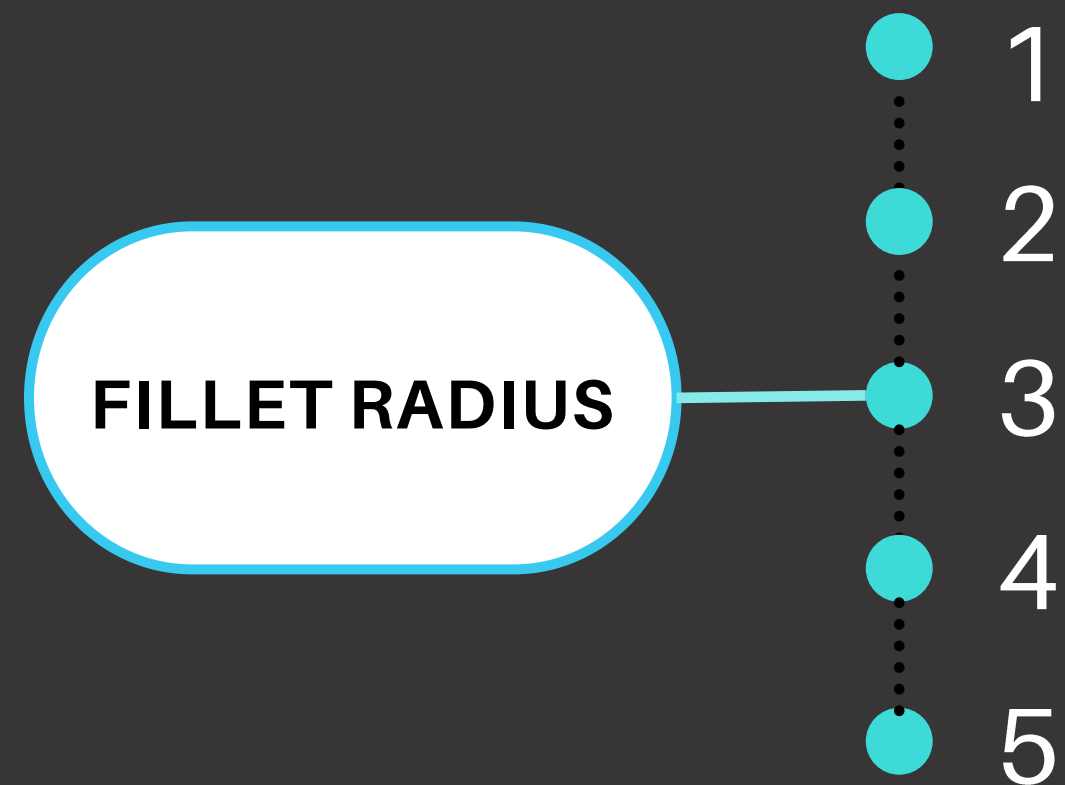
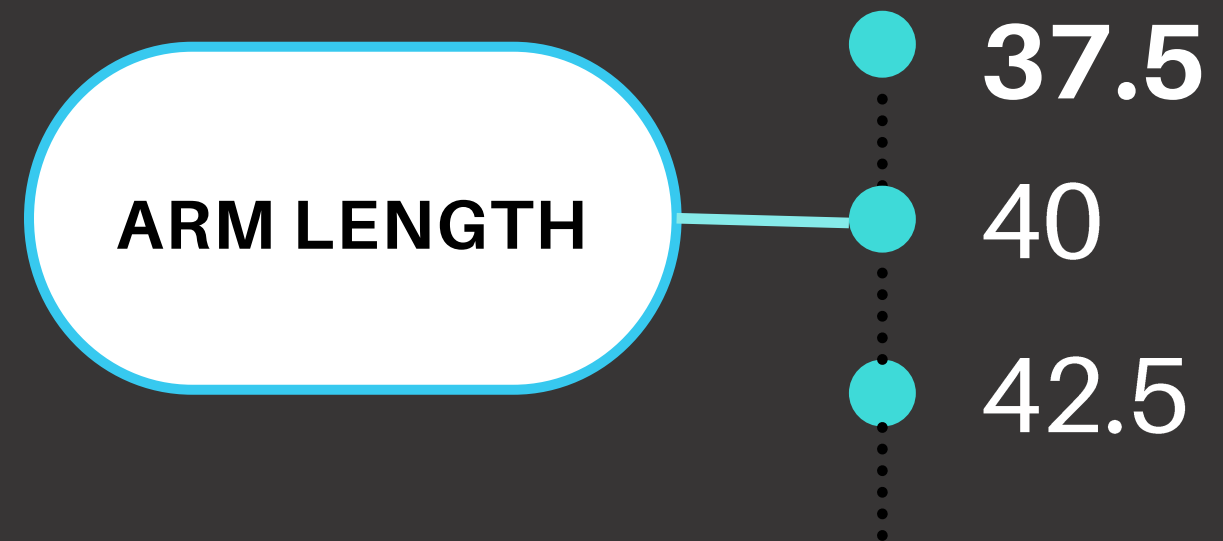
Inner Diameter

0.201mm =

Thickness

Actuator Contact  
diameter (OACD)

Fillet Radius



| PARAMETERS    |      |           |             |    |             |
|---------------|------|-----------|-------------|----|-------------|
| Arm Length    | 42.5 | 40.0      | <b>37.5</b> | -  | -           |
| Arm Angle     | 7    | 8         | 9           | 10 | <b>10.5</b> |
| Inner Dia     | 55   | <b>60</b> | -           | -  | -           |
| Fillet Radius | 1    | 2         | 3           | 4  | 5           |
| OACD          | 4    | 5         | 6           | -  | -           |

# INFLUENCE MATRIX

| X   | Y   | O1  | O2  | O3  | O4  | O5  | O6  | O7  | O8  | Ce  | I1  | I2  | I3  | I4  | I5  | I6  | I7  | I8  |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... |

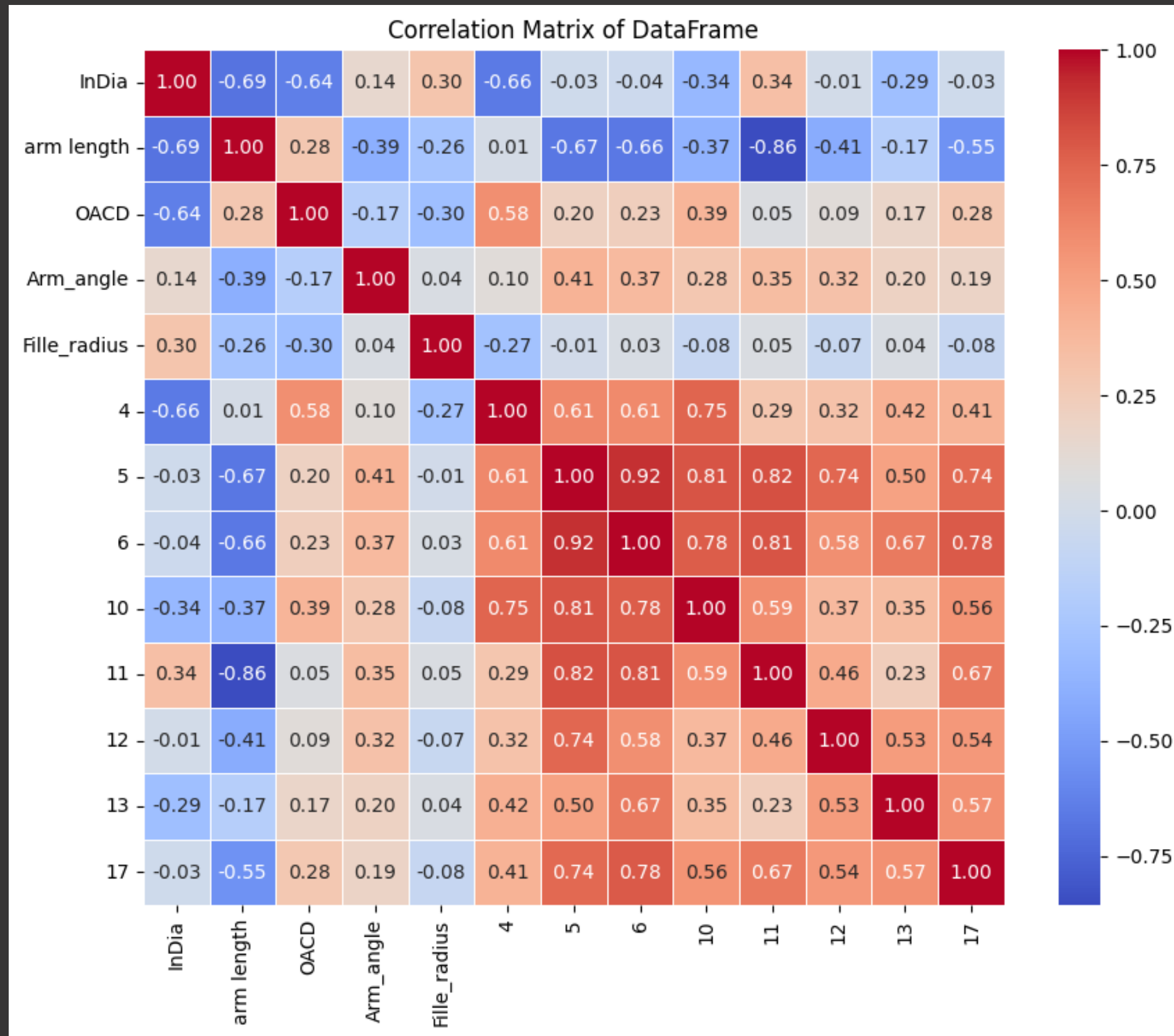
[CODE](#)

MATLAB CODE

[CODE](#)



# CORRELATION MATRIX

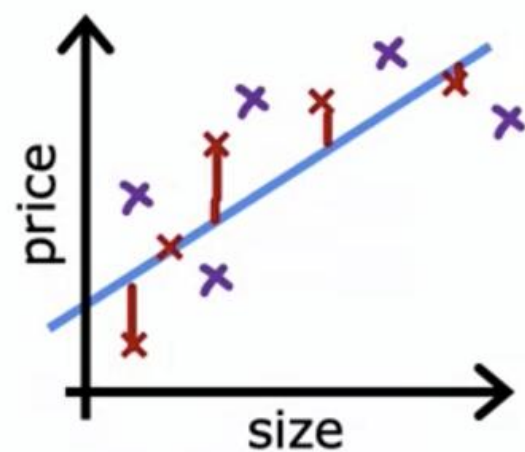


- Feature-Output Relations:
- InDia and arm length → strong negative correlation with several outputs.
- OACD and Arm\_angle → moderate positive correlation.
- Fille\_radius → very weak correlation (low importance).
- Output-Output Relations:
- 5, 6, 10, 11 → highly correlated (easy to predict together).
- 12 & 13 → weak correlations with inputs and other outputs → harder to predict.
- Takeaway: Strong dependencies among some outputs can be exploited with multi-output models, while weakly correlated outputs may need additional features/engineering.

# FIRST APPROACH :

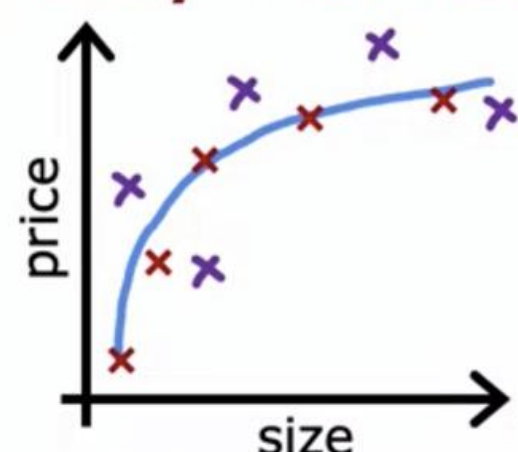
## RIDGE REGRESSION

### Bias/variance



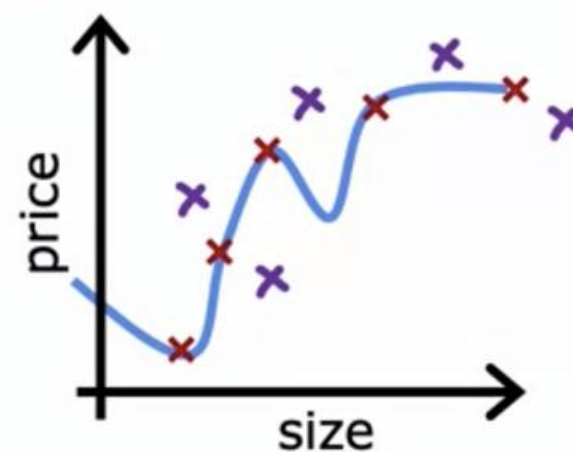
$$f_{\bar{w},b}(x) = w_1x + b$$

High bias  
(underfit)



$$f_{\bar{w},b}(x) = w_1x + w_2x^2 + b$$

"Just right"



$$f_{\bar{w},b}(x) = w_1x + w_2x^2 + w_3x^3 + w_4x^4 + b$$

High variance  
(overfit)

$d=1$   $\bar{J}_{\text{train}}$  is high  
 $\bar{J}_{\text{cv}}$  is high

$d=2$   $\bar{J}_{\text{train}}$  is low  
 $\bar{J}_{\text{cv}}$  is low

$d=4$   $\bar{J}_{\text{train}}$  is low  
 $\bar{J}_{\text{cv}}$  is high

Why Ridge Regression?

it minimizes variance by adding penalty to the overfitting.

it minimizes **bias + alpha(a) \* slope^2**

hence, sensitivity of the model reduces which prevents overfitting.

| <i>Model</i>      | <i>BIAS</i>        | <i>VARIANCE</i>    | <i>DIFFERENCE</i> |
|-------------------|--------------------|--------------------|-------------------|
| Elastic - net     | 41.384066          | 72.628185          | 31.2441           |
| Linear Regression | 57.32751951331115  | 89.2304366         | 31.9029           |
| Ridge Regression  | 58.267617821524766 | 91.59428673531225  | 33.3267           |
| Polynomial RR 2   | 26.663680          | 66.126302          | 39.462622         |
| Polynomial RR 3   | 17.174038016666618 | 106.49099166477806 | 89.3170           |
| Random Forest     | 8.932265061060601  | 59.60058374819435  | 50.6683           |

*Arm len*

*Arm Ang*

$$F_r$$

37.5

10

1

37.5

10

5

37.5

10

5

40.0

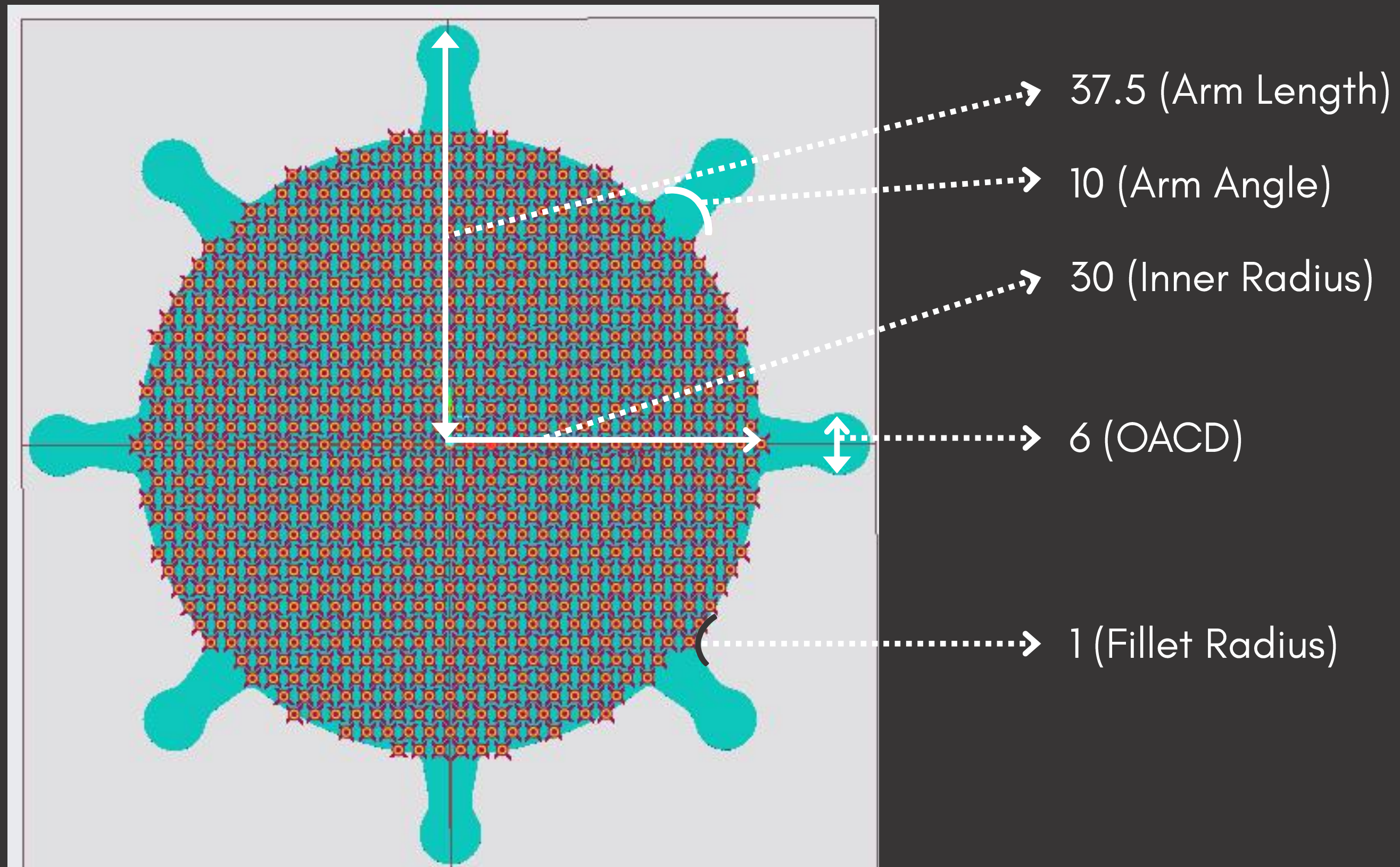
8

3

Elastic -net  
Ridge Regression  
Random Forest

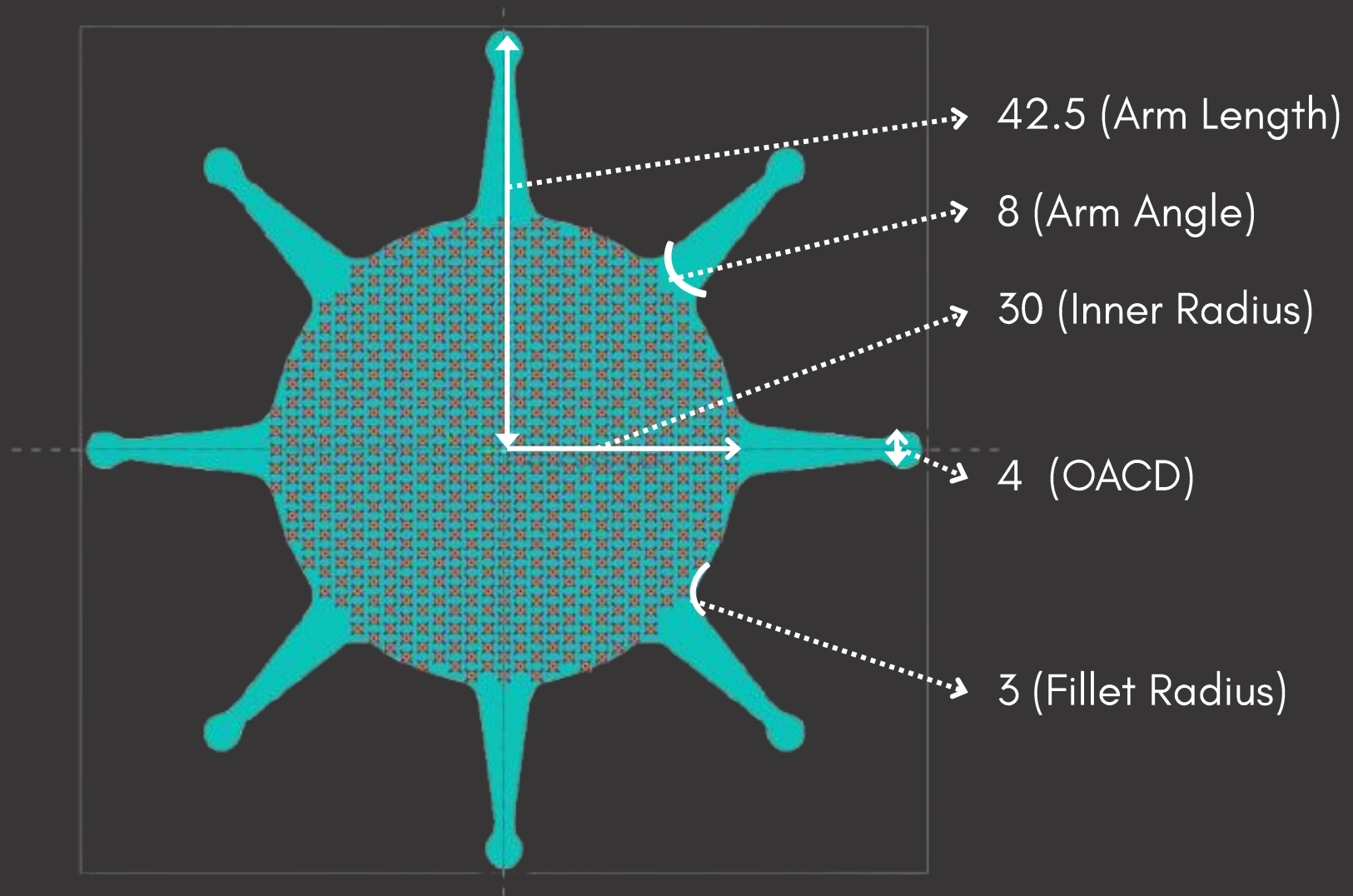
## Inverse Prediction with Optimization

- **Purpose:** Find input values (X) that produce a desired target output (Y\_target).
- **Objective Function:** Minimizes error =  $(\text{Model Prediction} - Y_{\text{target}})^2$ .
- **Optimization (scipy.minimize):**
  - Starts from an initial guess.
  - Adjusts inputs within set bounds.
  - Finds the X combination that best matches Y\_target.
- **Use Case:** Reverse-engineering the model → “What inputs give me this output?”



# FINAL SHAPE

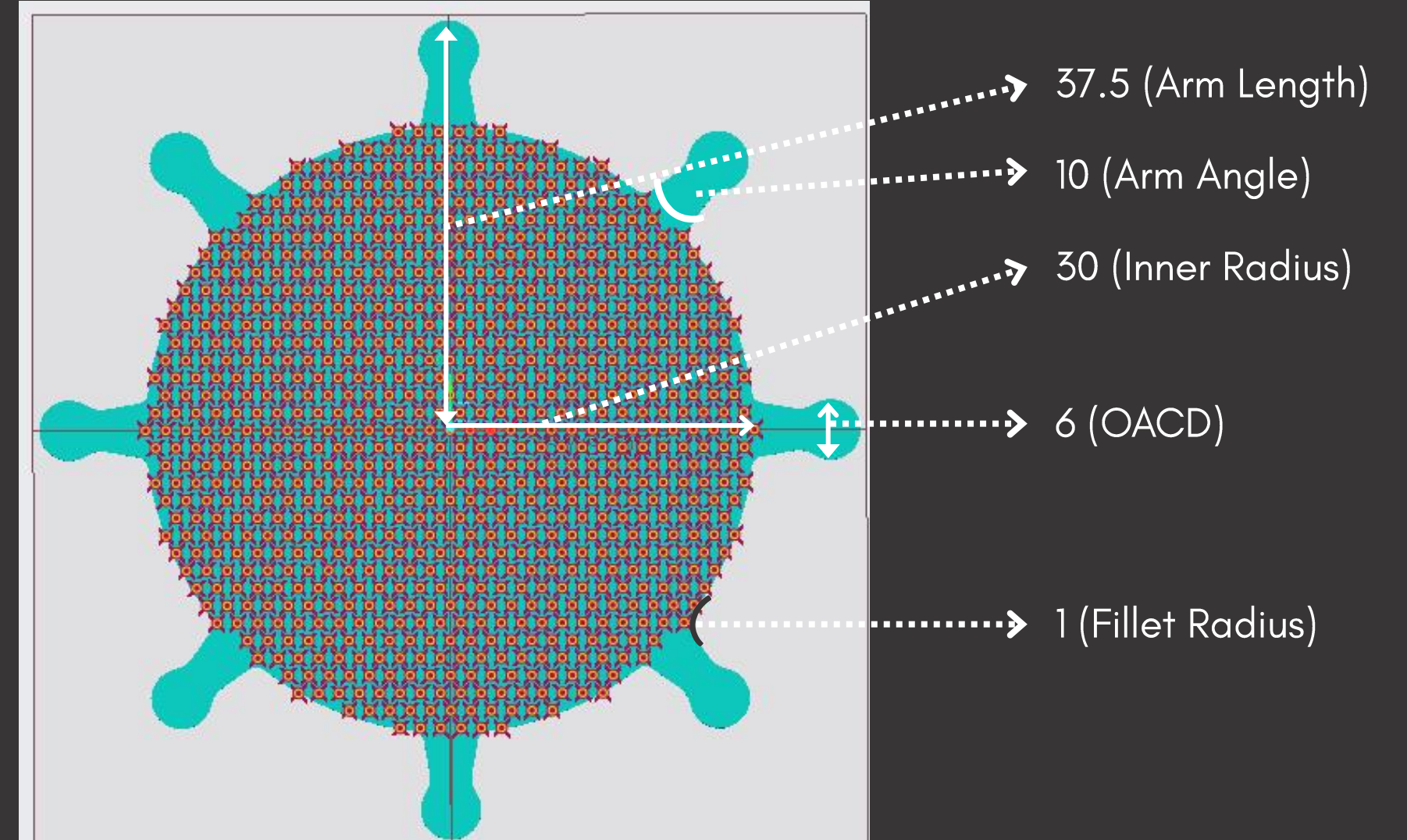
## INITIAL MODEL



### ACCURACIES :

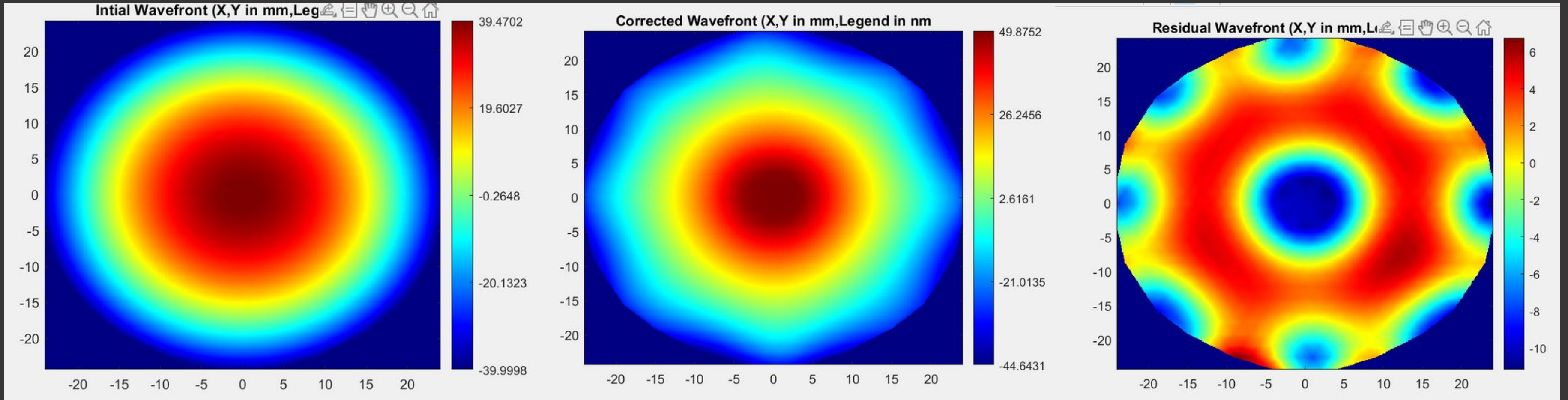
|         |         |        |      |
|---------|---------|--------|------|
| 68      | 72.9    | 71     | 65.6 |
| defocus | astig45 | astig0 | z=17 |

## ML Parameters

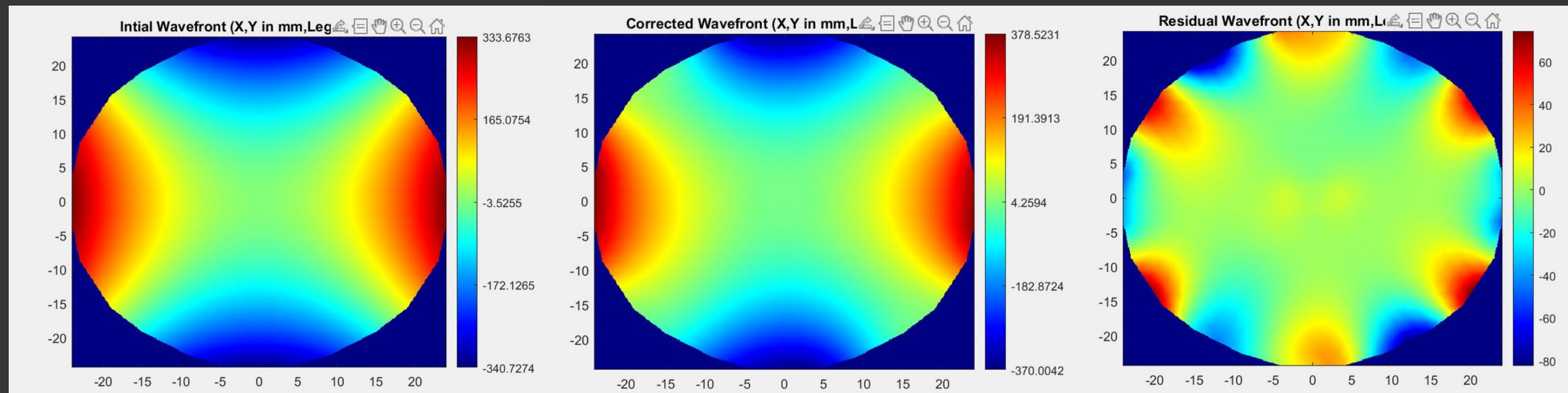


### ACCURACIES :

|         |         |        |       |
|---------|---------|--------|-------|
| 82.58   | 86.3    | 86.98  | 90.31 |
| defocus | astig45 | astig0 | z=17  |

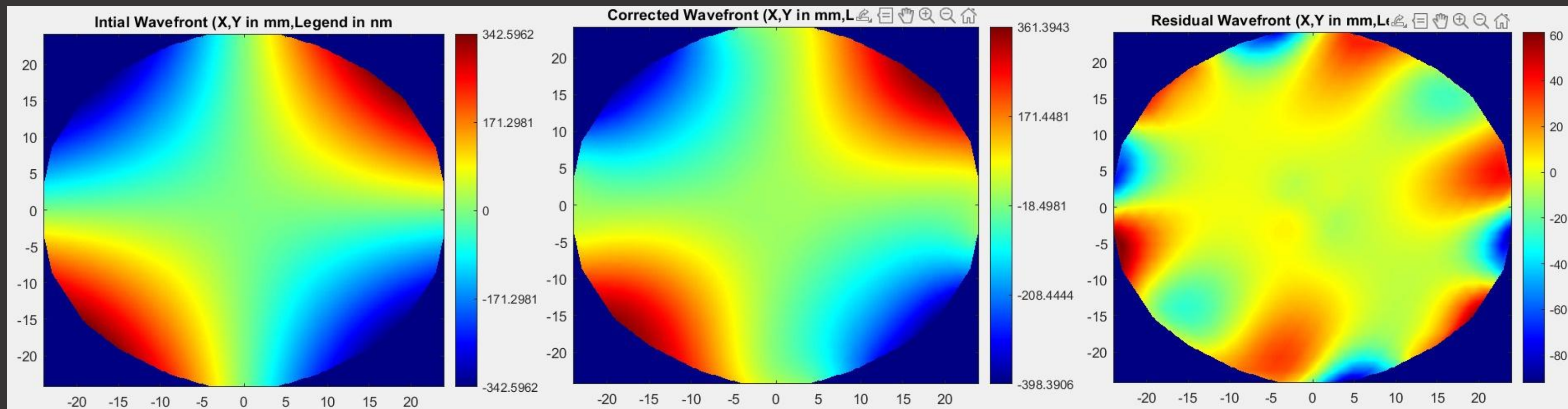


**DEFOCUS**  
**82.58**



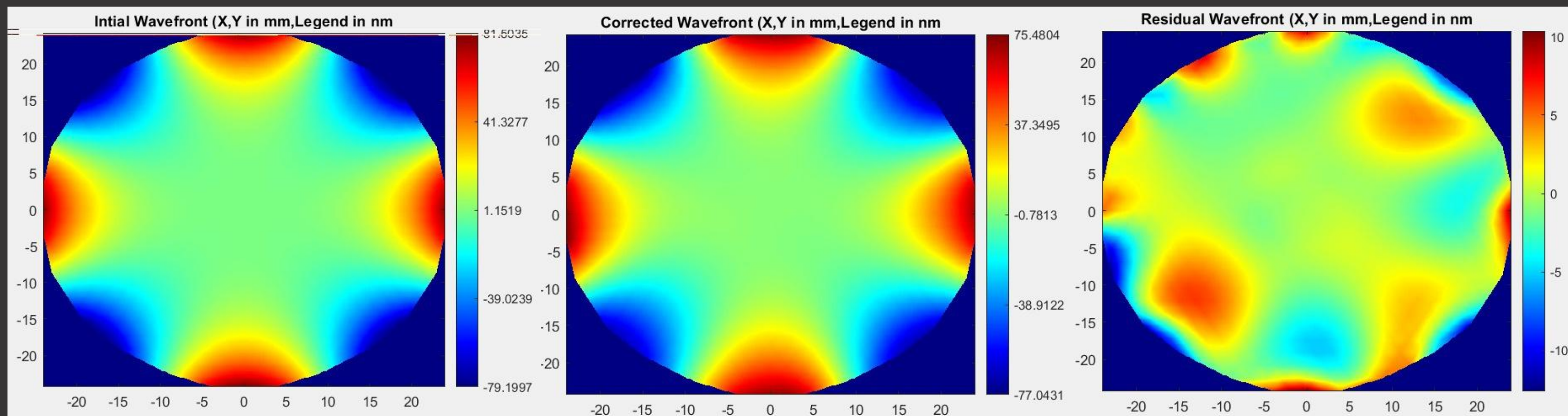
**ASTIG 45**

**86.3**



**ASTIG 0**

**86.98**



**Z =17**

**90.31**

**THANK YOU**